

Alex Lewandowski

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I design algorithms so that agents continue to develop their behaviour as they explore the world.

Education

Ph.D. in Computing Science

UNIVERSITY OF ALBERTA

2019 – Present

- » Supervisors: Dale Schuurmans, Marlos C. Machado
- » Research topics: continual learning, reinforcement learning, meta-learning
- » Anticipated graduation: 06/2026

M.Sc. in Statistics

UNIVERSITY OF ALBERTA

2016 – 2018

- » Supervisors: Ivor Cribben, Rohana Karunamuni
- » Thesis: Recurrent and Bayesian Kernel Learning for Small Data with Applications to Neuroimaging

Honours Bachelor in Mathematics

UNIVERSITY OF WATERLOO

2012 – 2016

- » Major: Mathematical Economics

Work Experience

Research Associate (Part-time), Noah's Ark Lab

HUAWEI TECHNOLOGIES CANADA CO., LTD.

2022 – 2023

- » Supervisor: Jun Luo
- » Developed a reinforcement learning framework for meta-learning to improve training speed of new agents

Research Associate (Full-time), Noah's Ark Lab

HUAWEI TECHNOLOGIES CANADA CO., LTD.

2021 – 2022

- » Supervisor: Jun Luo
- » Designed an evaluation methodology for generalization of reinforcement learning agents, applied to autonomous vehicles

Teaching Assistant

UNIVERSITY OF ALBERTA

2016 – 2018, 2019 – 2021

- » Computing Science: Facilitated discussion groups and developed assignments for a new reinforcement learning course. Provided mentorship and guidance for student projects in foundations of information retrieval
- » Mathematical & Statistical Sciences: Offered personalized support to students in all first and second-year classes at the Decima Robinson Support Centre. Conducted help sessions Statistics I/II, Regression Analysis and Time Series Analysis.

Papers & Preprints

Universal computation is intrinsic to language model decoding

01/2026

- » A. Lewandowski, M. C. Machado, D. Schuurmans
- » Preprint

The world is bigger: A computationally-embedded perspective on the big world hypothesis

12/2025

- » A. Lewandowski, A. Ramesh, E. Meyer, D. Schuurmans, M. C. Machado
- » Neural Information Processing Systems (NeurIPS spotlight)

Balancing expressivity and robustness: Constrained rational activations for reinforcement learning

08/2025

- » R. Surdej, M. Bortkiewicz, A. Lewandowski, M. Ostaszewski, C. Lyle
- » Conference on Lifelong Learning Agents (CoLLAs oral)

Plastic learning with deep Fourier features

04/2025

- » A. Lewandowski, D. Schuurmans, M. C. Machado
- » International Conference on Learning Representations (ICLR)

Learning continually by spectral regularization	04/2025
» A. Lewandowski , M. Bortkiewicz, S. Kumar, A. György, D. Schuurmans, M. Ostaszewski, M. C. Machado	
» International Conference on Learning Representations (ICLR)	
The need for a big world simulator: A scientific challenge for continual learning	08/2024
» S. Kumar, H. J. Jeon, A. Lewandowski , B. Van Roy	
» Finding The Frame at the Reinforcement Learning Conference (RLC workshop oral)	
Directions of curvature as an explanation for loss of plasticity	11/2023
» A. Lewandowski , H. Tanaka, D. Schuurmans, M. C. Machado	
» Preprint	
Reinforcement teaching	06/2023
» C. Muslimani*, A. Lewandowski *, D. Schuurmans, M. Taylor, J. Luo	
» Transactions on Machine Learning Research (TMLR)	
ZORB: A derivative-free backpropagation algorithm for neural networks	12/2020
» V. Ranganathan, A. Lewandowski	
» Beyond Backpropagation at Neural Information Processing Systems (NeurIPS workshop oral)	

Honors & Awards

Alberta Innovates Graduate Student Scholarship	2022—2024
University of Alberta, Department of Computing Science	
President's Doctoral Prize of Distinction	2021—2024
University of Alberta, Department of Computing Science	
NSERC Postgraduate Scholarships – Doctoral	2021—2024
University of Alberta, Department of Computing Science	
Josephine Mitchell Scholarship	2017—2018
University of Alberta, Department of Mathematical and Statistical Sciences	
Profiling Alberta's Graduate Students Award	2017
University of Alberta, Department of Mathematical and Statistical Sciences	
Queen Elizabeth II Graduate Scholarship	2016
University of Alberta, Department of Mathematical and Statistical Sciences	
President's Scholarship	2012
University of Waterloo	

Community

Organizer	RLC 2025 (Local Chair), Finding the Frame Workshop at RLC 2025, Openmind Research Institute Banff Retreat 2023-2024, ICLR SSBM Social 2021, NeurIPS SSBM Social 2020, ICLR Reinforcement Learning Social 2020
Reviewer	NeurIPS 2021-2026 (Top Reviewer: 2022, 2025), ICLR 2020-2026 (Outstanding Reviewer: 2021, 2022), ICML 2021-2024 (Outstanding Reviewer: 2022), RLC 2024-2026, CoLLAs 2025-2026
Other	Academic Director at Computing Science Graduate Student Association 2022—2023

*Equal contribution